Support for IT Project Management

Topic 3

Comparative Analysis of System Modeling Software

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Presentation Structure

Part I

Functionality Specification

1. User Needs
2. Software Functionality (MoSCoW Model)
3. Use Case Diagram

Part II

Comparative Analysis

1. Software Under Analysis
2. Comparison Matrix
3. Critical Analysis

Part I

Functionality Specification

User Needs & Context

User Need

N1	**Standardized Project Initiation**: Start designs using pre-defined architectural templates.
N2	**Intuitive Modeling Workspace**: A friendly UI with rapid-access palettes for all necessary UML tools.
N3	**Dynamic Canvas Manipulation**: Ability to resize, move, and stylize (color/text) model elements freely.
N4	**Design Accuracy & Validation**: Ensure the model follows technical rules and catches design mistakes automatically
N5	**Real-Time Collaboration**: Work simultaneously with live cursors and shared editing.
N6	**Contextual Communication**: Integrated chat or issue-tracking for design decisions.
N7	**Data Persistence**: Automated saving of model changes to prevent data loss.
N8	**Automated Code Creation**: Turn diagrams into initial code skeletons to speed up work.
N9	**Documentation Export**: Generate shareable technical files in PNG, PDF, or SVG.
N10	**Version History**: Track changes, compare versions, and audit the design evolution.

Functionality Catalog (1/2) - Must & Should Have

ID	Functionality	Input	Output	Description	MoSCoW
F1	**Intuitive Modeling Canvas**	User gestures (drag/drop, palette)	UML diagram file	A friendly UI with a specialized palette and an auto-layout engine to keep diagrams organized. Responds to N2 and N3.	M
F2	**Full UML Notation Set**	Diagram type selection	Specialized notation set	Support for all 14 UML 2.x diagram types to visualize system behavior and structure. Responds to N2.	M
F3	**Multi-Format Export**	UML model + target format	PNG / PDF / SVG file	High-resolution exports for technical documentation and sharing with stakeholders. Responds to N6.	M
F4	**Model Error Detection**	UML model elements	Validation report	Automatic check for design contradictions and notation errors against UML rules. Responds to N4	S

M = Must Have

S = Should Have

C = Could Have

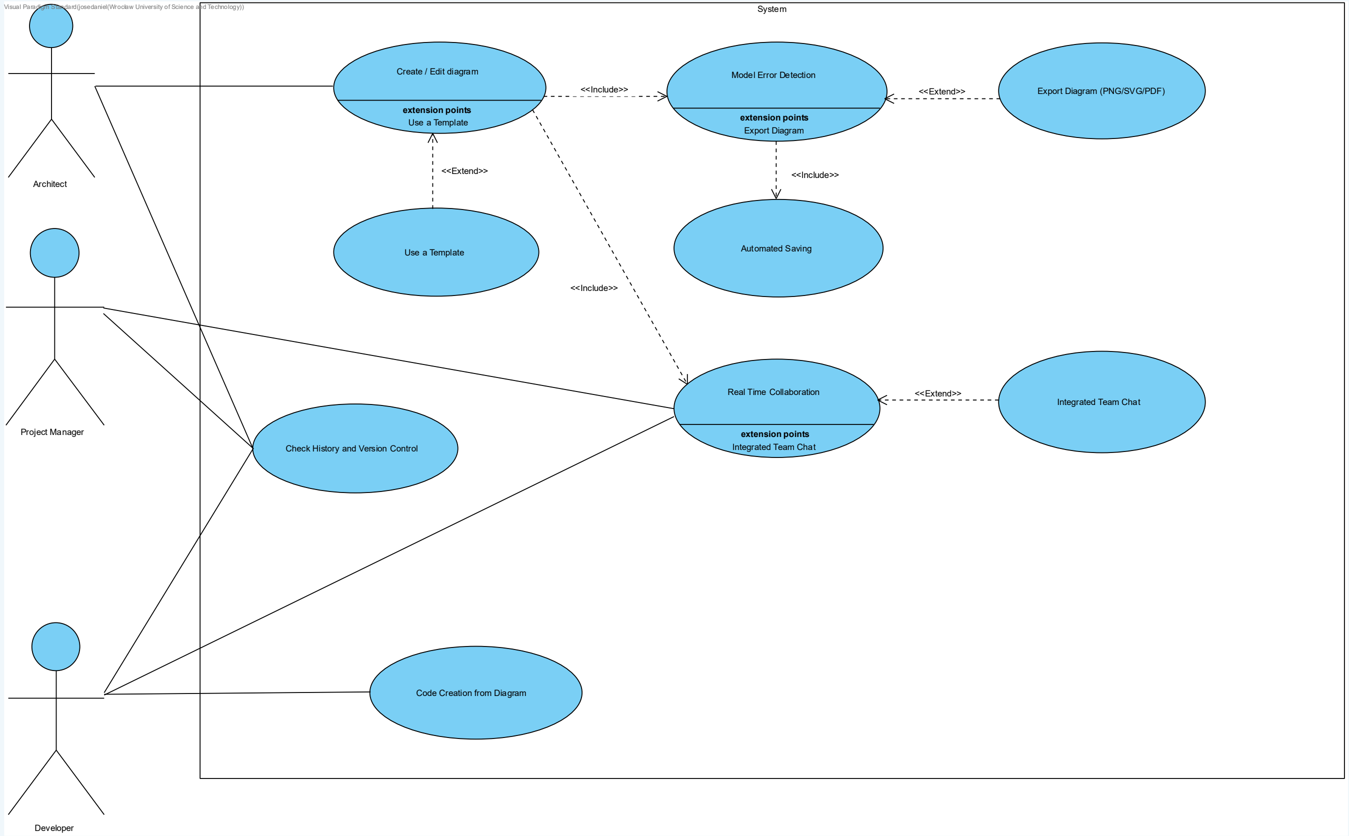
W = Won't Have

Functionality Catalog (2/2) - Should, Could & Won't Have

ID	Functionality	Input	Output	Description	MoSCoW
F5	**Real-Time Online Collaboration**	Concurrent user actions	Synchronized model	Allows multiple users to work simultaneously with live cursors and shared editing. Responds to N5.	S
F6	**Automated Saving**	Model change events	Persistent data state	Background saving of all changes to ensure no data is lost during sessions. Responds to N7.	S
F7	**Automated Code Creation**	UML Diagrams	Source code skeletons	Automatically generates initial code structures from diagrams to speed up development. Responds to N8.	C
F8	**Integrated Team Chat**	User messages / comments	Discussion feed	Built-in communication tools to discuss design decisions and issues in context. Responds to N9.	C
F9	**History & Version Tracking**	Version snapshots	Comparison report	Tracks the evolution of the model, allowing users to compare versions and audit changes. Responds to N10.	C
F10	**Architectural Templates**	Template selection	Pre-built structure	Pre-defined patterns to help users start projects quickly. Categorized as Won't Have to focus on core tools. Responds to N1.	W

M = Must Have   S = Should Have   C = Could Have   W = Won't Have

UML Use Case Diagram - System Modeling Software



Actors: System Analyst, Software Architect, Developer | 9 Use Cases | Relationships: <<include>>, <<extend>>, Association

Part II

Comparative Analysis of Available Solutions

Software Under Analysis

#	Tool Name	Type	Producer / Provider	Country	Licence
1	**Enterprise Architect**	Standalone	Sparx Systems	Australia	Commercial
2	**Visual Paradigm**	Web + Standalone	Visual Paradigm Int'l	Hong Kong	Commercial / Free tier
3	**Lucidchart**	Web Application	Lucid Software Inc.	USA	Commercial / Free tier
4	**StarUML**	Standalone	MKLab	South Korea	Commercial (formerly open-source)
5	**draw.io (diagrams.net)**	Web Application	JGraph Ltd / diagrams.net	UK	Free / Open-source
6	**Modelio**	Standalone	Modeliosoft / Softeam	France	Free / Open-source

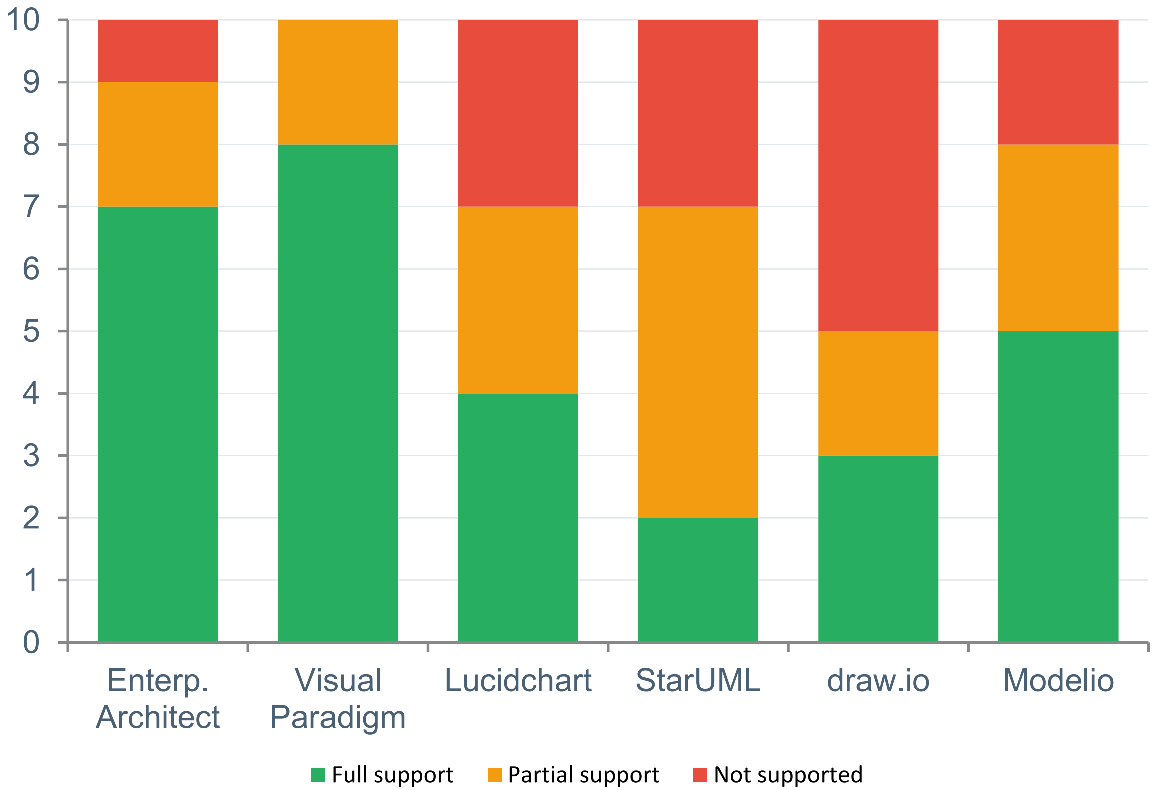
Selection criteria: active maintenance (2024-2026), UML 2.x compliance, publicly accessible, representative of market diversity (commercial, freemium, open-source)

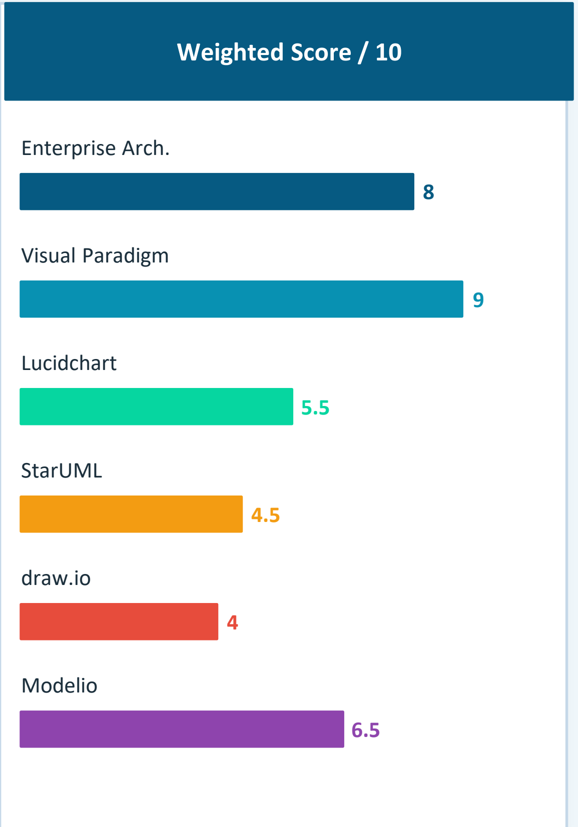
Comparison Matrix - Functionalities vs. Tools

Functionality \ Tool	Enterprise Architect	Visual Paradigm	Lucidchart	StarUML	draw.io	Modelio
F1 Intuitive Modeling Canvas	Full	Full	Full	Full	Partial	Full
F2 Full UML Notation Set (all 14 UML 2.x)	Full	Full	Partial	Full	None	Full
F3 Multi-Format Export (PNG / PDF / SVG)	Full	Full	Partial	Partial	Partial	Full
F4 Model Error Detection / Validation	Full	Full	None	Partial	None	Full
F5 Real-Time Online Collaboration	Partial	Full	Full	None	Full	None
F6 Automated Saving	Full	Full	Full	Partial	Full	Partial
F7 Automated Code Creation	Full	Full	None	Partial	None	Full
F8 Integrated Team Chat	None	Partial	Partial	None	None	None
F9 Version Control Integration	Full	Partial	None	None	None	Partial
F10 Template & Pattern Library	Partial	Full	Full	Partial	Full	Partial

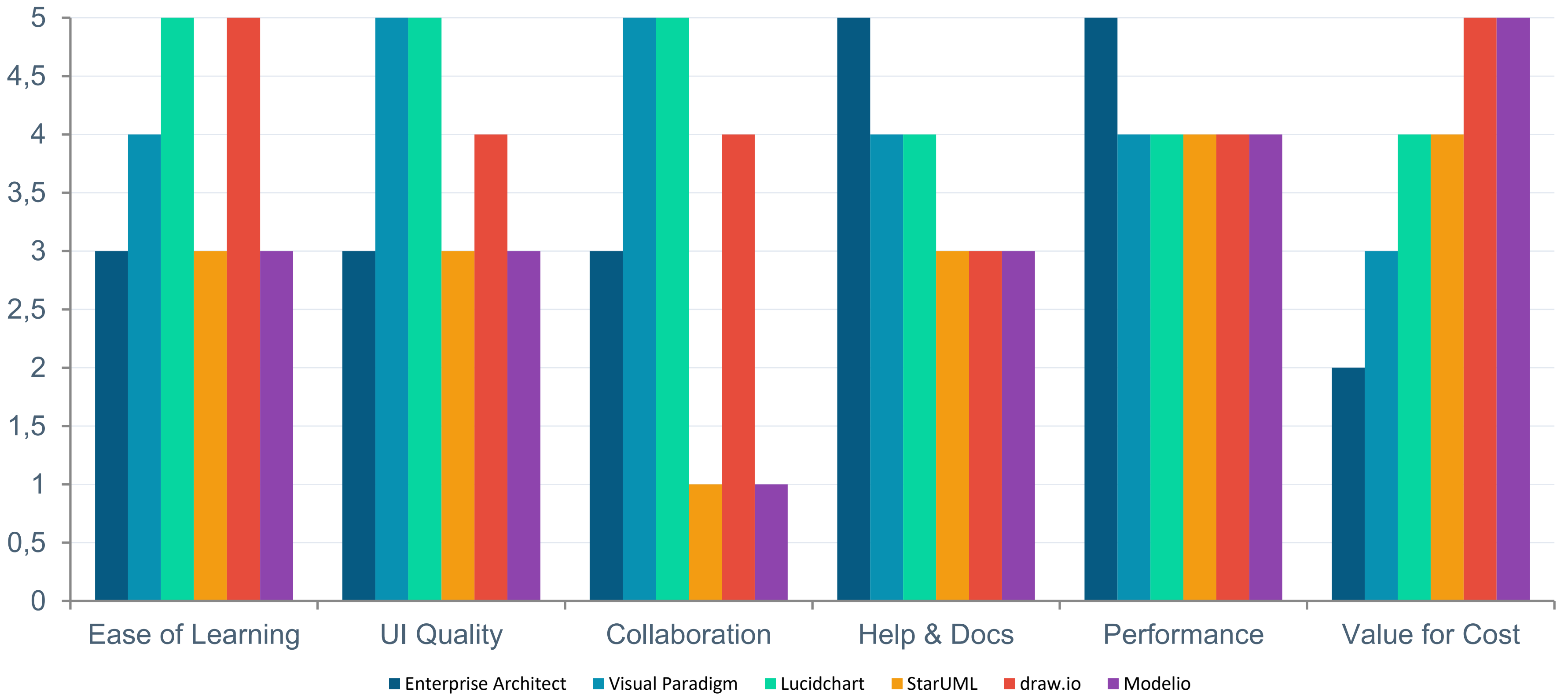
Full = Full support   Partial = Partial support   None = Not supported

Feature Coverage - Weighted Score per Tool





Usability Analysis - 6 Dimensions (Scale 1-5)



Scale: 1 (Poor) to 5 (Excellent) | Sources: G2, Capterra, official documentation, hands-on evaluation | Scores are representative averages

Critical Analysis - Strengths & Limitations

Enterprise Architect

+ Most complete UML feature set; excellent code gen & reverse eng; mature ecosystem

- Steep learning curve; expensive licensing; outdated UI design

StarUML

+ Free base version; lightweight; good for individual modelling tasks

- Many features partial or plugin-dependent; no collaboration support

Visual Paradigm

+ Best balance of features + usability; strong collaboration; modern web interface

- Subscription model adds up; advanced features require higher tiers

draw.io (diagrams.net)

+ Completely free; offline-capable; excellent integrations (GitHub, Confluence)

- Not a true UML tool; no model semantics, validation, or code generation

Lucidchart

+ Outstanding ease of use; best real-time collaboration; seamless integrations

- Limited UML compliance; no code gen or reverse eng; no XMI export

Modelio

+ Free & open-source; full UML + BPMN support; strong Java code generation

- Complex interface; no built-in collaboration; smaller community

Summary & Recommendations

Rank	Tool	Score	Key Justification
1st	**Visual Paradigm**	9.0/10	Best overall balance of features, usability, and collaboration support
2nd	**Enterprise Architect**	8.0/10	Full professional UML suite; best for large enterprise projects & code traceability
3rd	**Modelio**	6.5/10	Top free/open-source choice for full UML compliance and code generation
4th	**Lucidchart**	5.5/10	Ideal for quick diagrams and collaboration; not for formal UML modeling
5th	**StarUML**	4.5/10	Good lightweight free tool; limited for team or enterprise use cases
6th	**draw.io**	4.0/10	Best free general diagramming; does not meet UML standard semantics

Recommendation: For formal system modeling in IT projects, **Visual Paradigm** offers the best cost-benefit ratio (web access, full UML, collaboration). For budget-constrained teams, **Modelio** provides full UML capability at zero cost.

Thanks for your Attention!

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